PROGRAM - 3

TITLE: Code a dockerized python Flask or Node.js application

Project Structure

Program-3

- app.py
- requirements.txt
- Dockerfile

Step-1: Create a Dockerfile

```
1RV24MC018_Ananya_h@user-ThinkPad-E480:~$ cd labprograms
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/labprograms$ mkdir program-3
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/labprograms$ cd program-3
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/labprograms/program-3$ nano requirements.txt
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/labprograms/program-3$ nano app.py
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/labprograms/program-3$ nano Dockerfile
```

```
  GNU nano 6.2
#use python
FROM python:latest

#set up the working directory
WORKDIR /app

#copy and install the necessary dependencies
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

#copy the source application
COPY . .

#expose the port
EXPOSE 5000

#commands to run
CMD ["python","app.py"]
```

Step-2: Create a requirements.txt file

```
  GNU nano 6.2
Flask==2.3.3
```

Step3: Create a app.py file

```
 GNU nano 6.2
from flask import Flask
app=Flask(__name__)
@app.route("/")
def home():
    return "Hello, from flask application"
if __name__=='__main__':
    app.run(host='0.0.0.0', port=5000)
```

Step-4: To build an image

> docker build -t prg-3 .

```
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/labprograms/program-3$ docker build -t prg-3 .
[+] Building 1.2s (10/10) FINISHED                                                              docker:default
 => [internal] load build definition from Dockerfile                                                      0.0s
 => => transferring dockerfile: 346B                                                                      0.0s
 => [internal] load metadata for docker.io/library/python:latest                                          1.0s
 => [internal] load .dockerignore                                                                         0.0s
 => => transferring context: 2B                                                                           0.0s
 => [1/5] FROM docker.io/library/python:latest@sha256:1ad1a43b5e2478e62056bbc28028afd858185d73bf4d6a439cbb058b6800a96d  0.0s
 => [internal] load build context                                                                         0.0s
 => => transferring context: 283B                                                                         0.0s
 => CACHED [2/5] WORKDIR /app                                                                             0.0s
 => CACHED [3/5] COPY requirements.txt .                                                                  0.0s
 => CACHED [4/5] RUN pip install --no-cache-dir -r requirements.txt                                       0.0s
 => [5/5] COPY . .                                                                                        0.0s
 => exporting to image                                                                                    0.0s
 => => exporting layers                                                                                   0.0s
 => => writing image sha256:3c7c092252339105a52fc7d6f6c241c747eda3a29e739d8edc36f3b922dce62c              0.0s
 => => naming to docker.io/library/prg-3                                                                  0.0s
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/labprograms/program-3$ docker images
REPOSITORY    TAG       IMAGE ID       CREATED          SIZE
prg-3         latest    3c7c09225233   6 seconds ago    1.13GB
<none>        <none>    ea6814133d0f   2 minutes ago    1.13GB
prg-2         latest    e6f2e5ab289f   6 days ago       137MB
prg1          latest    80e1235b9aa3   6 days ago       132MB
hello-world   latest    1b44b5a3e06a   3 months ago     10.1kB
```

Step-5: To run

> docker run -p 5000:5000 prg-3

```
c^C1RV24MC018_Ananya_h@user-ThinkPad-E480:~/labprograms/program-3$ docker run -p 5000:5000 prg-3
 * Serving Flask app 'app'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on all addresses (0.0.0.0)
 * Running on http://127.0.0.1:5000
 * Running on http://172.17.0.2:5000
Press CTRL+C to quit
172.17.0.1 - - [07/Nov/2025 04:02:20] "GET / HTTP/1.1" 200 -
172.17.0.1 - - [07/Nov/2025 04:02:21] "GET /favicon.ico HTTP/1.1" 404 -
172.17.0.1 - - [07/Nov/2025 04:02:34] "GET / HTTP/1.1" 200 -
172.17.0.1 - - [07/Nov/2025 04:02:35] "GET / HTTP/1.1" 200 -
172.17.0.1 - - [07/Nov/2025 04:02:35] "GET / HTTP/1.1" 200 -
172.17.0.1 - - [07/Nov/2025 04:02:37] "GET / HTTP/1.1" 200 -
172.17.0.1 - - [07/Nov/2025 04:02:40] "GET / HTTP/1.1" 200 -
^C1RV24MC018_Ananya_h@user-ThinkPad-E480:~/labprograms/program-3$ docker ps -a
CONTAINER ID   IMAGE     COMMAND            CREATED          STATUS                     PORTS     NAMES
e2e76458448e   prg-3     "python app.py"    7 minutes ago    Exited (0) 5 seconds ago             keen_wing
4f4f8c678cf6   prg-3     "python app.py"    17 minutes ago   Exited (0) 7 minutes ago             trusting_lamarr
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/labprograms/program-3$ docker rm e2e
e2e
```

Step-6: Check in your browser and open http://172.17.0.2:5000